



**Patuakhali Science and Technology University**  
Faculty of Computer Science and Engineering

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**CCE 310 :: Software Development Project-I**  
Sessional Project Proposal

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**Project title: Medcore**  
Submission Date :

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# Project Proposal: Medicores - A Cross-Platform Medical Management Application

## 1. Abstract

Medicores is a cross-platform Flutter-based medical appointment and e-prescription app designed for both patients and doctors. It provides appointment scheduling, video consultations, digital prescriptions, and patient history tracking with integrated payment support for a seamless healthcare experience. In an age where efficient healthcare access and management are crucial, Medicores aims to provide a simplified and powerful interface to manage and retrieve valuable medical resources from anywhere.

## 2. Objectives

- Simplify appointment booking and management for patients and doctors.
- Enable secure e-prescription and comprehensive medical history tracking.
- Integrate secure payment options for online/offline consultation fees.
- Improve communication via in-app chat and real-time notifications.
- To offer a seamless, secure, and intuitive experience across Android, iOS, and potentially web platforms.

## 3. Problem Statement

Existing solutions for medical appointment and health record management are often fragmented, platform-specific, or lack comprehensive features. Patients frequently struggle with:

- **Fragmented Medical Records:** Health information scattered across multiple providers, clinics, and paper files, making it difficult to maintain a complete history.
- **Inefficient Appointment Scheduling:** Manual booking processes, long wait times, and difficulties in rescheduling or canceling appointments.
- **Lack of Digital Prescriptions:** Reliance on paper prescriptions, leading to potential loss, illegibility, and inefficiencies in refills.
- **Limited Communication:** Challenges in direct, secure communication between patients and doctors outside of consultations.
- **Payment Hurdles:** Inconvenient payment methods for consultations, often requiring physical presence.

There is a clear need for a comprehensive, user-friendly tool that centralizes medical management, streamlines processes, and enhances communication for both patients and healthcare providers.

## 4. Related Work

Several existing applications address parts of the medical management ecosystem, but often lack a unified approach or specific features:

- **Telehealth Platforms:** Offer video consultations but may lack robust personal health record management or e-prescription capabilities.
- **Hospital/Clinic-Specific Apps:** Often tied to a single healthcare provider, limiting comprehensive management across multiple doctors.
- **Medication Reminder Apps:** Focus solely on medication adherence, without integrating appointments or records.
- **Personal Health Record Apps:** May allow record storage but lack direct appointment booking or consultation features.

Medicore aims to address these limitations by offering a more comprehensive, integrated, and user-centric solution.

## 5. Scope

Medicore will be developed as a cross-platform application using Flutter, allowing it to run on Android, iOS, and potentially web browsers. The initial version will focus on providing a rich set of features for managing appointments, e-prescriptions, and medical records. It will also include secure user authentication, role-based dashboards, and integrated payment processing.

Later on, it can be extended to include advanced features like integration with wearable health devices, AI-powered symptom analysis, deeper EHR integrations, and more sophisticated analytics for both patients and doctors.

## 6. Methodology

The development of Medicore will follow an agile methodology, allowing for iterative improvements and user feedback.

## 6.1. Technology Stack

- **Frontend:** Flutter (Dart) for cross-platform mobile development.
- **Backend:** Firebase (for authentication, real-time database, cloud functions) and/or PHP + MySQL (for custom logic and relational data).
- **Database:** Firestore (for NoSQL flexibility with Firebase) or MySQL (for structured relational data).
- **Payment Gateway:** SSLCommerz for secure and localized payment processing.
- **Notifications:** Firebase Cloud Messaging (FCM) for real-time alerts and reminders.
- **UI/UX Design:** Figma for wireframing, mockups, and interactive prototypes.

## 6.2. Design Principles

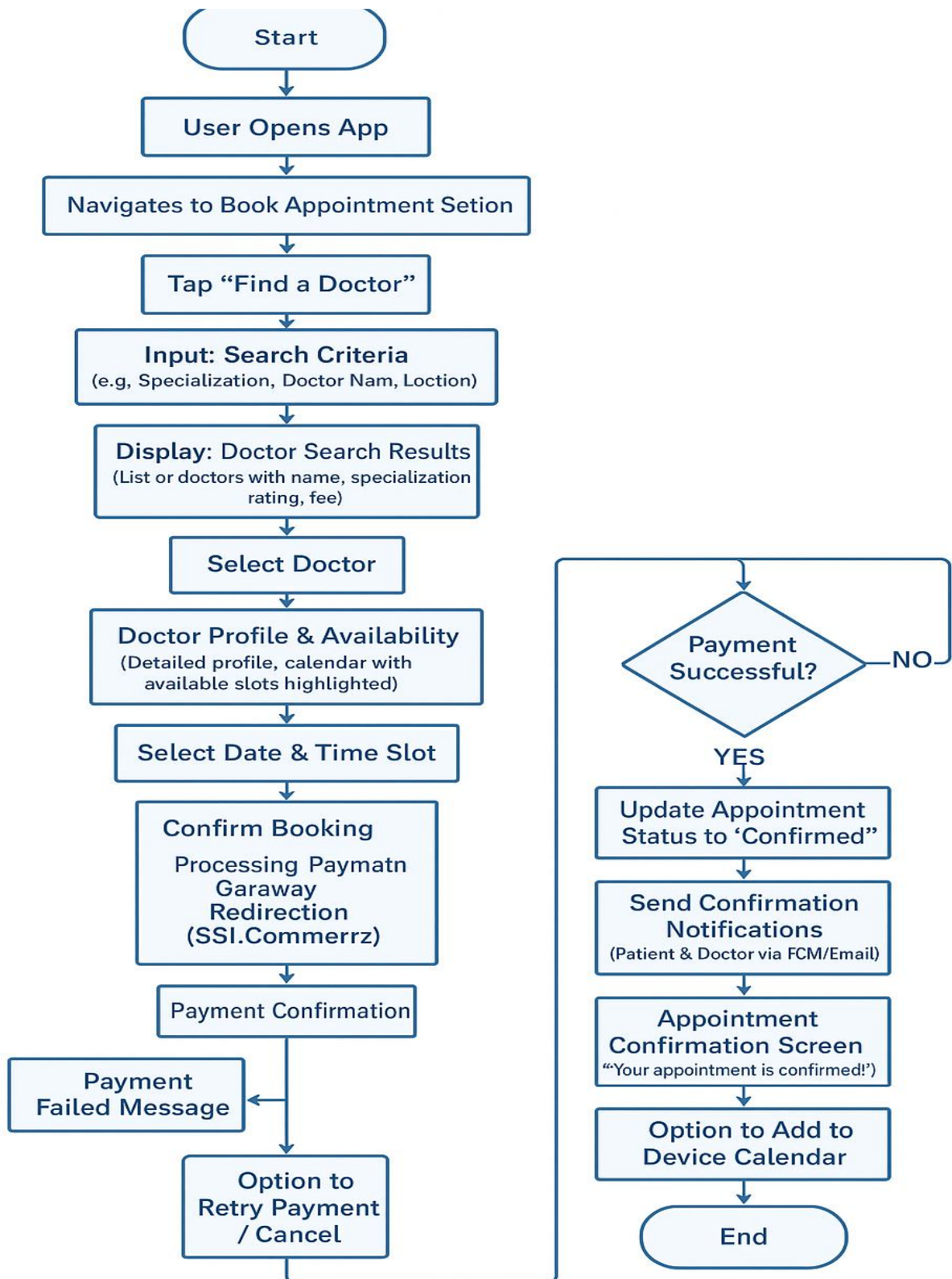
The design of Medcore will adhere to the following principles:

- **User-Centric Design:** Focusing on intuitive navigation, clear UI elements, and ease of use for diverse user groups (patients, doctors, admins).
- **Material Design / Cupertino Design:** Following platform-specific guidelines for a consistent and modern UI/UX on Android and iOS, respectively.
- **Responsive Design:** Ensuring the application works seamlessly on various screen sizes and orientations.
- **Security & Privacy by Design:** Integrating robust security measures and adhering to healthcare data privacy regulations from the outset.
- **Accessibility:** Designing for users with varying abilities, including considerations for screen readers, font sizing, and clear visual hierarchies.

## 7. Visual Models

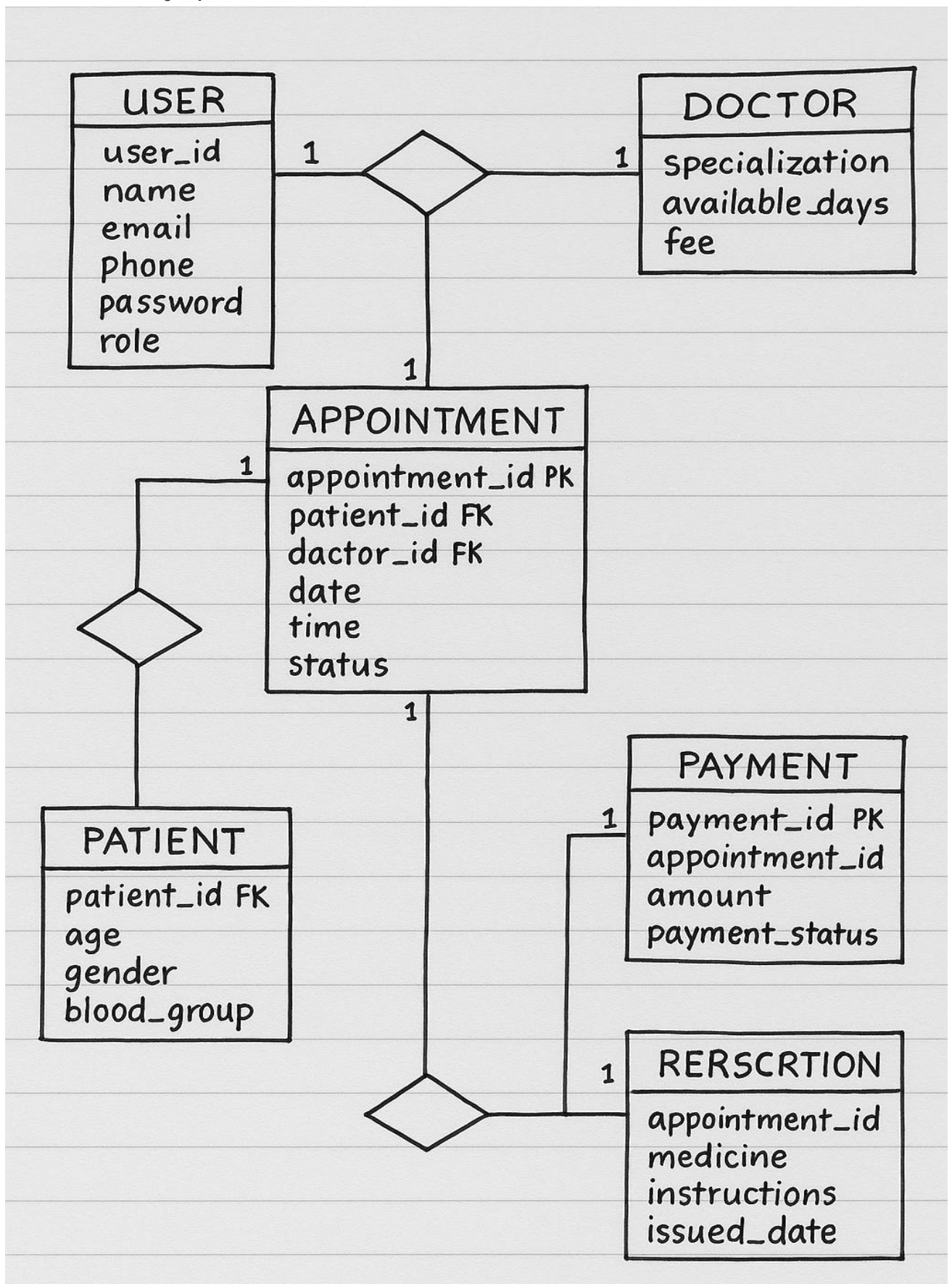
This section provides conceptual descriptions of the essential visual and structural models that would underpin the "Medcore" application's design and development. As an AI, I cannot generate actual visual diagrams or interactive UI mockups, but these descriptions serve as blueprints.

### 7.1. Flow Chart Diagram



**Fig: Flow chart Diagram**

## 7.2. Entity Relationship (ER) Diagram



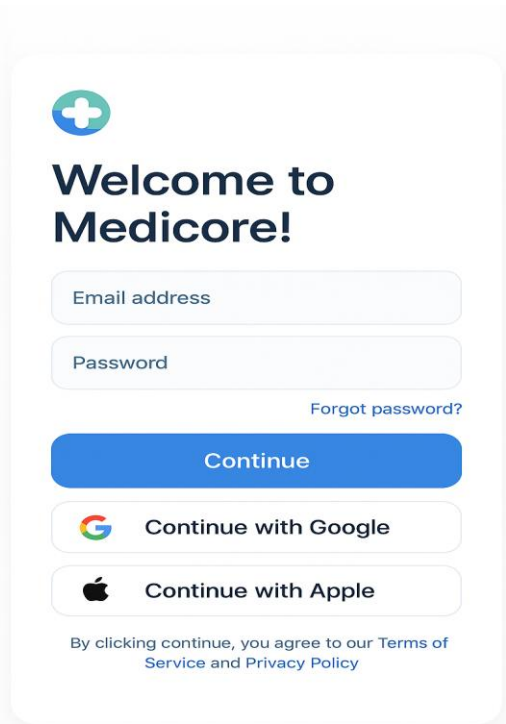
7.3. Timeline (Gantt Chart)

| Phase                               | Task Name                         | Duration (Weeks) | Dependencies | Milestones            |
|-------------------------------------|-----------------------------------|------------------|--------------|-----------------------|
| Phase 1: Discovery & Planning       |                                   |                  |              |                       |
|                                     | 1.1 Requirements Gathering        | 2                | -            | Requirements Sign-off |
|                                     | 1.2 Technical Architecture Design | 3                | 1.1          | Architecture Approved |
| Phase 2: UI/UX Design & Prototyping |                                   |                  |              |                       |
|                                     | 2.1 Wireframing & User Flows      | 2                | 1.2          | Wireframes Approved   |
|                                     | 2.2 High-Fidelity UI Design       | 4                | 2.1          | UI Designs Approved   |
|                                     | 2.3 Interactive Prototyping       | 3                | 2.2          | Prototype Ready       |
| Phase 3: Core MVP Development       |                                   |                  |              |                       |
|                                     | 3.1 Backend & API                 | 12               | 1.2          | Backend Alpha         |

|                                                 |                                              |   |                     |                     |
|-------------------------------------------------|----------------------------------------------|---|---------------------|---------------------|
|                                                 | Development                                  |   |                     |                     |
|                                                 | 3.2 Frontend Dev (Auth, Dashboards)          | 6 | 2.3, 3.1 (start)    | User Auth Ready     |
|                                                 | 3.3 Frontend Dev (Doc Listing, Appt Booking) | 8 | 3.2, 3.1 (progress) | Booking Flow Ready  |
|                                                 | 3.4 Frontend Dev (E-Prescription, Records)   | 6 | 3.3, 3.1 (progress) | Core Features Ready |
|                                                 | 3.5 Payment Gateway Integration              | 4 | 3.1, 3.3            | Payments Integrated |
|                                                 | 3.6 Notifications Integration                | 2 | 3.1, 3.3            | Notifications Live  |
| <b>Phase 4: Testing &amp; Quality Assurance</b> |                                              |   |                     |                     |
|                                                 | 4.1 Functional & Integration Testing         | 4 | 3.6                 | QA Cycle 1 Complete |
|                                                 | 4.2 Security & Performance Testing           | 3 | 4.1                 | Security Audit      |
|                                                 | 4.3 Bug Fixing & Refinement                  | 3 | 4.1, 4.2            | UAT Ready           |
|                                                 | 4.4 User Acceptance                          | 2 | 4.3                 | UAT Sign-off        |

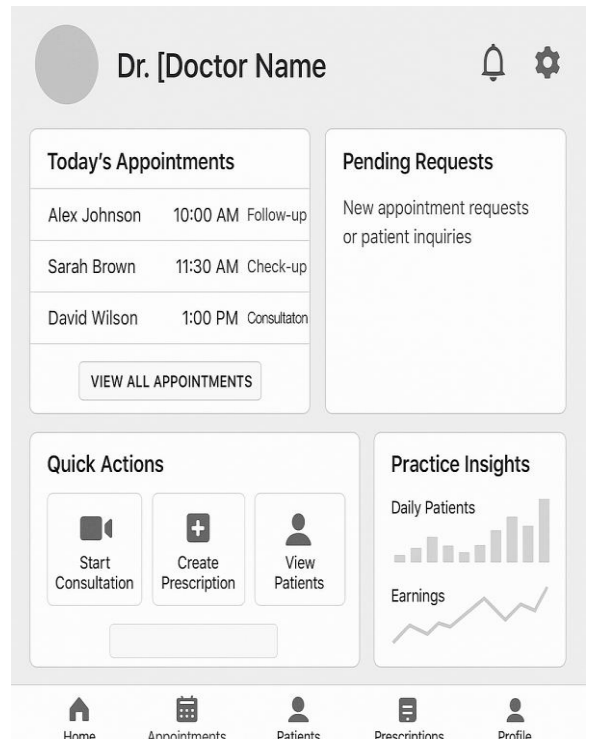
|                                             |                                       |         |                           |                  |
|---------------------------------------------|---------------------------------------|---------|---------------------------|------------------|
|                                             | Testing (UAT)                         |         |                           |                  |
| <b>Phase 5: Deployment &amp; Launch</b>     |                                       |         |                           |                  |
|                                             | 5.1 App Store Submission              | 2       | 4.4                       | App Live         |
|                                             | 5.2 Initial Marketing & PR            | 3       | 4.4 (parallel)            | Launch Campaign  |
| <b>Phase 6: Post-Launch &amp; Iteration</b> |                                       |         |                           |                  |
|                                             | 6.1 Monitoring & Support              | Ongoing | 5.1                       | Continuous       |
|                                             | 6.2 Feature Enhancements (v1.1, etc.) | Ongoing | User Feedback & Analytics | Updates Released |

## 7.4. UI Mockups



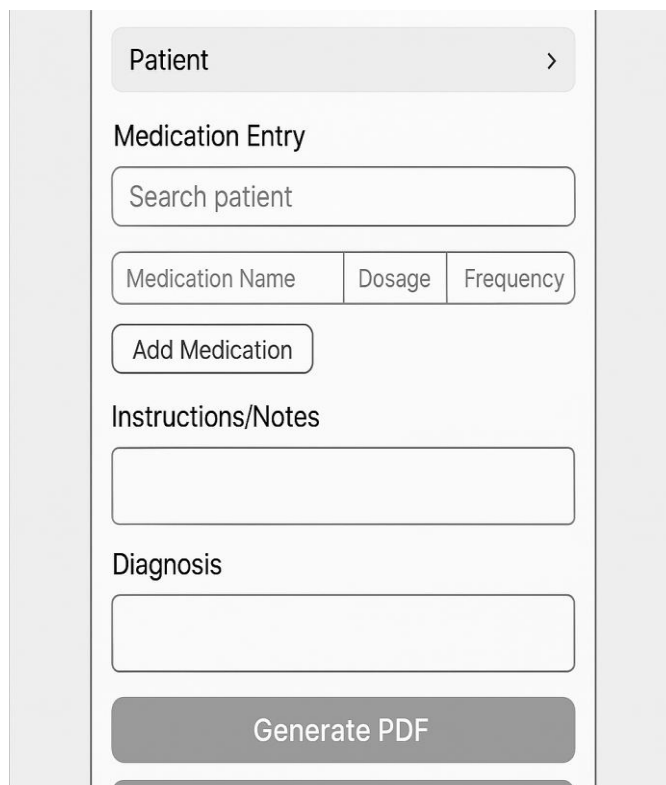
UI Mockup of the Medicore login page. The page features a green cross logo at the top left. Below the logo, the text "Welcome to Medicore!" is displayed in a large, bold font. Underneath, there are two input fields for "Email address" and "Password". A link "Forgot password?" is positioned to the right of the password field. A blue "Continue" button is located below the input fields. Further down, there are two buttons for social login: "Continue with Google" and "Continue with Apple". At the bottom, a small text line states: "By clicking continue, you agree to our Terms of Service and Privacy Policy".

**Fig: UI Mockups Logging page**



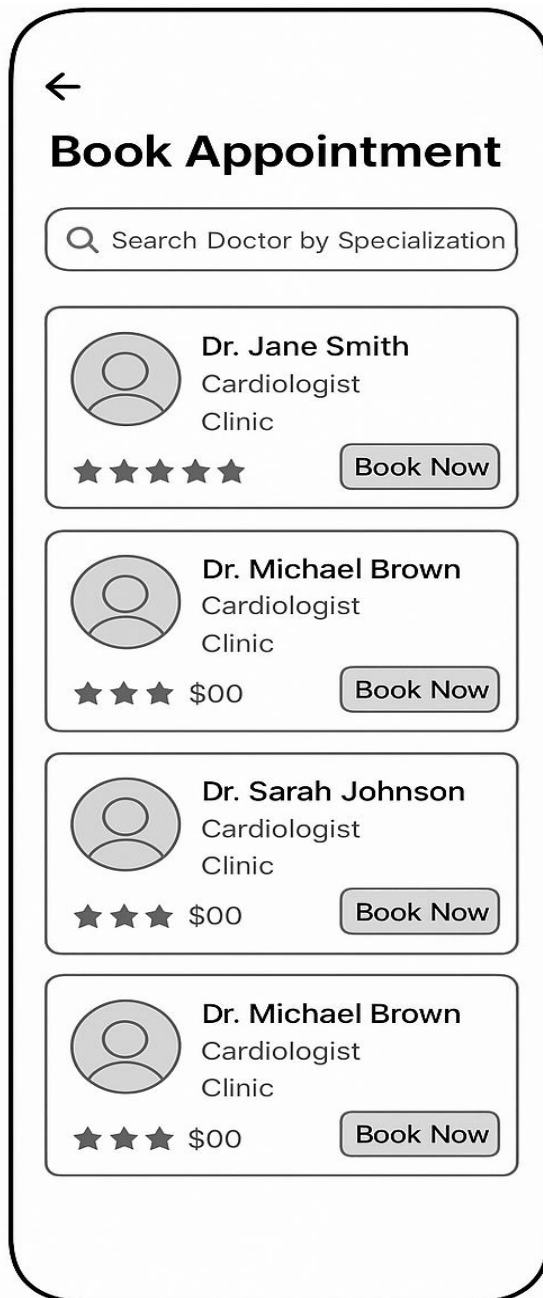
UI Mockup of the Doctor Dashboard. The dashboard is for "Dr. [Doctor Name]" and includes a notification bell and a settings gear icon. It is divided into several sections: "Today's Appointments" (listing Alex Johnson at 10:00 AM Follow-up, Sarah Brown at 11:30 AM Check-up, and David Wilson at 1:00 PM Consultation, with a "VIEW ALL APPOINTMENTS" button), "Pending Requests" (stating "New appointment requests or patient inquiries"), "Quick Actions" (with buttons for "Start Consultation", "Create Prescription", and "View Patients"), and "Practice Insights" (showing "Daily Patients" as a bar chart and "Earnings" as a line chart). A bottom navigation bar contains icons for Home, Appointments, Patients, Prescriptions, and Profile.

**Fig: Doctor Dashboard**

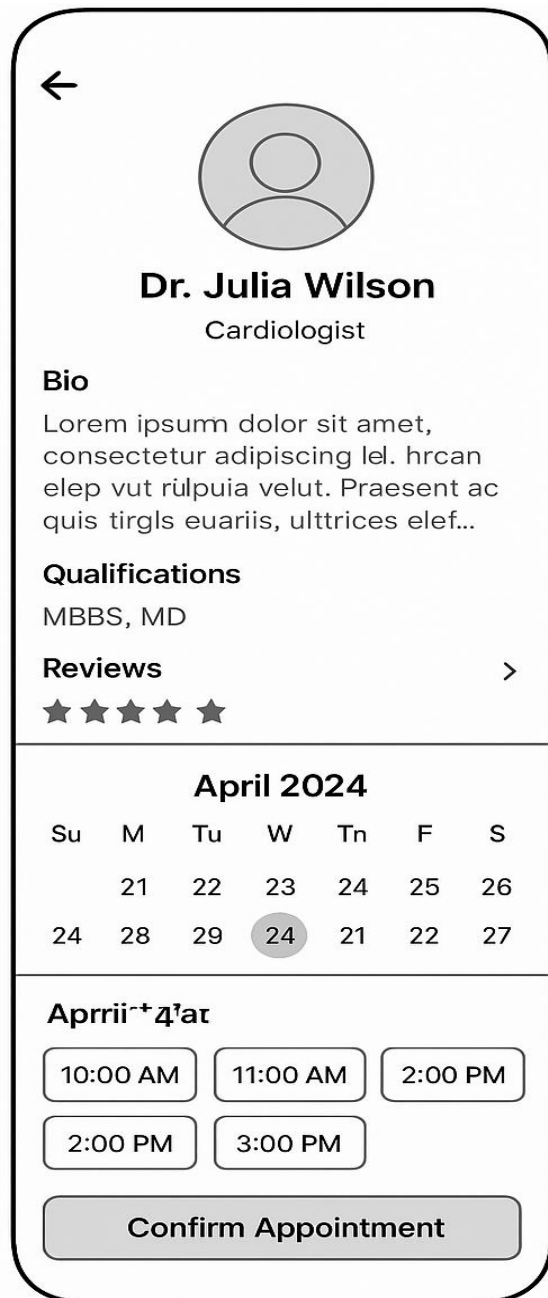


UI Mockup of the Medicine sector form. The form is titled "Patient" with a right arrow. Below the title is the "Medication Entry" section, which includes a "Search patient" input field, a table with columns "Medication Name", "Dosage", and "Frequency", and an "Add Medication" button. The "Instructions/Notes" section has a large text input field. The "Diagnosis" section also has a large text input field. At the bottom, there is a "Generate PDF" button.

**Fig: Medicine sector**



Search Doctor by Specialization or Name



April 24 April 24

## 8. Limitations

1. **Browser Extensions:** Currently, browser extensions are not planned, but the application will be designed to allow easy integration with popular browsers in the future for potential web-based access.
2. **Initial Feature Set:** The initial version will focus on core functionalities (appointment, e-prescription, basic records, payments) and may not include all advanced features like deep EHR integration, AI

diagnostics, or comprehensive wearable device syncing. These will be considered for future iterations.

3. **Offline Functionality:** While core data will be synced, extensive offline capabilities (e.g., full medical record access without internet) might be limited in the initial MVP.

## 9. Result

The expected outcome of Medcore is a fully functional, secure, and cross-platform application that allows users to manage their personal healthcare efficiently and empowers doctors to streamline their practice. It aims to improve patient engagement, reduce administrative burdens, and foster better communication within the healthcare ecosystem.

## 10. Security and Compliance

Given the sensitive nature of medical data, security and compliance will be paramount throughout the development lifecycle. This includes:

- **Data Encryption:** Implementing robust encryption for data at rest and in transit (e.g., SSL/TLS for communication, database encryption).
- **Authentication & Authorization:** Strong user authentication mechanisms (e.g., Firebase Email OTP, multi-factor authentication) and granular role-based access control to ensure users only access data relevant to their role.
- **Regular Security Audits:** Conducting periodic security assessments, vulnerability scanning, and penetration testing by third-party experts.
- **Regulatory Adherence:** Ensuring strict compliance with relevant healthcare data privacy regulations such as HIPAA (for the US market), GDPR (for the EU market), and local data protection laws in Bangladesh. Legal consultation will be sought to ensure full adherence.
- **Secure API Design:** Implementing secure API endpoints with proper input validation, rate limiting, and secure token management to prevent unauthorized access and data breaches.
- **Audit Trails:** Maintaining comprehensive audit logs for all data access and modification activities.

## 11. Conclusion

Medcore aims to be a transformative digital solution for personal healthcare management. By combining a user-friendly interface built with Flutter, a robust backend, and essential features like appointment booking, e-prescriptions, and payment integration, it will empower patients and streamline operations for doctors. Our unwavering commitment to security and compliance will ensure a trustworthy platform, making Medcore an indispensable tool for a more connected, efficient, and patient-centric healthcare ecosystem.

## 12. Weekly Report

| No. | Date | Objective                         | Remarks |
|-----|------|-----------------------------------|---------|
| 1   |      | Project proposal with initial R&D |         |
| 2   |      |                                   |         |
| 3   |      |                                   |         |
| 4   |      |                                   |         |
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